CURRICULUM VITAE

THOMAS ROSS GRAY

Email: thomasgray@brandeis.edu | ORCID: 0000-0002-6176-005X

CITIZENSHIP

United States of America, NA
Colombia, SA

ACADEMIC HISTORY & TRAINING

Ph.D. Neuroscience August 2020 – September 2026
Brandeis University
Waltham, MA

B.S. Neuroscience August 2016 – May 2020
Georgia State University
Atlanta, GA

The Genome Access Course November 2019
Cold Spring Harbor Laboratories
Cold Spring Harbor, NY

Fundamentals in Fluorescence Microscopy Course Jan 2026
Brandeis University Microscopy Core
Waltham, MA

RESEARCH EXPERIENCE

Doctoral Candidate – R01 Diversity Supplement Awardee Oct 2023 – August 2025
NIH T32 Training Grant August 2020 – August 2022
Principal Investigator: Donald B. Katz
Brandeis University, Waltham, MA
Visiting Scholar with Joost Maier, Ph.D. (Wake Forest University, January 2023)

Doctoral Rotation Student - NIH T32 Training Grant August 2020 – May 2021
Rotations with Paul Garrity, Ph.D.; Eve Marder Ph.D.; and Leslie Griffith Ph.D.
Brandeis University, Waltham, MA

Undergraduate Assistant Researcher - NIH IMSD R25 Fellow March 2017 – May 2020
Principal Investigator: Daniel N. Cox, Ph.D.
Georgia State University, Atlanta, GA

Undergraduate Assistant Researcher - Volunteer June 2019 – September 2019
Principal Investigator: Manfred Schmidt, Ph.D.
Georgia State University, Atlanta, GA

COVID-19 Epidemiology Assistant July 2020 – September 2020
Supervisor: Amanda Seaman, MPH
Georgia Department of Public Health, Decatur, GA

AWARDS, HONORS, AND FUNDING

R01 Diversity Supplement Awardee ($154,216) 2023
Co-PI Jetstream2 ACCESS Allocation (400,000 SUs and 12TB Storage) 2023, 2024

Association for Chemoreception Sciences Conference Travel Award ($2,400)	2022, 2023
NIH R13 SACNAS NDiSTEM Conference Travel Award ($1,200)	2022
Volen Retreat Best Poster Prize ($50)	2021
NIH T32 Predoctoral Training Grant ($51,156)	2020-2022
NIH R25 IMSD Fellowship, Initiative for Maximizing Student Development ($35,000)	2017-2020
The Symposium for Undergraduate Research Exploration at UT Austin Travel Award ($1200)	2019
Admission to Honors College	2017
HOPE Scholarship	2017
Dean's List	2016, 2018

PUBLICATIONS

1. Himmel NJ, Sakurai A, Letcher JM, Patel AA, Maggie N. Benson MN, Bhattacharjee S, **Gray TR**, Cymbalyuk GS, and Cox DN. (2023) Chloride-dependent mechanisms of multimodal sensory discrimination and neuropathic sensitization *eLife* https://doi.org/10.7554/eLife.76863

2. Himmel NJ, Letcher JM, Sakurai A, **Gray TR**, Benson MN, Donaldson KJ, and Cox DN. (2021). Identification of a neural basis for cold acclimation in *Drosophila* larvae, *iScience* https://doi.org/10.1016/j.isci.2021.102657

3. Himmel NJ, **Gray TR**, Cox DN. (2020). Phylogenetics identifies two eumetazoan TRPM clades and an 8th TRP family, TRP soromelastatin (TRPS), *Molecular Biology and Evolution* 37, 7, (2034-2044), https://doi.org/10.1093/molbev/msaa065

4. Himmel NJ, Letcher JM, Sakurai A, **Gray TR**, Benson MN, and Cox DN. (2019). *Drosophila* menthol sensitivity and the Precambrian origins of transient receptor potential-dependent chemosensation. *Philosophical Transactions of the Royal Society B.* 374(1785):20190369, https://doi.org/10.1098/rstb.2019.0369

PREPRINTS

1. Himmel NJ, Sakurai A, Letcher JM, Patel A, Bhattacharjee S, Benson MN, **Gray TR,** Cymbalyuk GS, and Cox DN. Chloride-dependent mechanisms of multimodal sensory discrimination and neuropathic sensitization in *Drosophila, bioRxiv*. doi: https://doi.org/10.1101/2021.12.10.472093

2. Himmel NJ, Letcher JM, Sakurai A, **Gray TR**, Benson MN, Donaldson KJ, and Cox DN. The evolution of cold nociception in drosophilid larvae and identification of a neural basis for cold acclimation, *bioRxiv*. doi: doi.org/10.1101/2021.01.04.425280

3. Himmel NJ, **Gray TR**, and Cox DN. (2019). Phylogenetic inference identifies two eumetazoan TRPM clades and an 8th family of TRP channel, TRP soromelastatin (TRPS). *bioRxiv*. doi: 10.1101/860445

4. Himmel NJ, Letcher JM, Sakurai A, **Gray TR**, Benson MN, and Cox DN. (2019). *Drosophila* menthol sensitivity and the Precambrian origins of TRP-dependent chemosensation. *bioRxiv*. doi: 10.1101/690933

ORAL PRESENTATIONS

1. **Gray TR**, Vincis R. Multimodality in Chemosensory Systems. May 2025. *ACHEMS Career Networking Seminar Series,* Online

2. Katz DB, **Gray TR**. Functional dependence of insular-piriform coupling during chemosensory processing. June 2024. *International Symposium on Olfaction and Taste,* Reykjavik, Iceland

3. **Gray TR,** Katz DB. The relationship of gustatory and olfactory activity during retronasal olfaction. April 2023. *ACHEMS Annual Conference,* Fort Myers, FL.

4. **Gray TR**, Himmel NJ, Cox DN. Phylogenetics identifies TRP soromelastatin (TRPS), an 8th TRP family. April 2020. *GSURC Research Conference,* Atlanta, GA.

5. **Gray TR**, Himmel NJ, Cox DN. The neuropeptide Ccap is required in multimodal sensory neurons in order to drive stimulus-appropriate behavior. November 2019. *Annual Symposium for Undergraduate Research Exploration at the University of Texas at Austin*, Austin, TX.

6. **Gray TR**, Himmel NJ, Cox DN. Neuropeptides function in multimodal sensory neurons in order to drive stimulus-appropriate behavior selection. April 2019. *Oral presentation at the National Council for Undergraduate Research*, Atlanta, GA.

7. **Gray TR**. Dissecting the molecular and neural circuit bases of nociception. October 2017. *Course-based Undergraduate Research Experience Showcase*, Atlanta, GA.

PRESENTATIONS AND PUBLISHED ABSTRACT

1. **Gray TR,** Craddock A, Katz DB. April 2026 *ACHEMS Annual Conference,* St. Petersburg, FL.

2. **Gray TR,** Mahmood A, Craddock A, Goldstein I, White A, Katz DB. Neural Responses to Retronasal Stimuli in Rodents. September 2025. *Volen Center for Complex Systems Retreat*, West Dennis, MA.

3. White A, **Gray TR**, Katz DB. Somatosensory Influence on Retronasal Odor Perception. August 2025. *SciFest,* Waltham, MA

4. **Gray TR,** Mahmood A, Craddock A, Goldstein I, White A, Katz DB. Gustatory-Olfactory Cortical Interactions in Response to Multimodal Stimuli. April 2025 *ACHEMS Annual Conference,* Fort Myers, FL.

5. **Gray TR,** Goldstein I, Katz DB. Temporal dynamics of gustatory-olfactory cortical processing in rats. October 2024. *Society for Neuroscience Conference,* Chicago, IL

6. Surendran A, Goldstein I, **Gray TR**, Maigler KM, Katz DB. Optogenetic Inhibition of Gustatory Cortex Disrupts Taste-Potentiated Odor Associations. October 2024. *Society for Neuroscience Conference*, Chicago, IL

7. White A, **Gray TR**, Katz DB. Taste Preference Before and After Surgery Remain Stable in Rats. August 2024. *SciFest,* Waltham, MA

8. Surendran A, Goldstein I, **Gray TR**, Maigler KM, Katz DB. Disrupting Taste Cortex Prevents Taste-Odor Learning Potentiation. August 2024. *SciFest,* Waltham, MA

9. Goldstein I, Surendran A, **Gray TR**, Maigler KM, Katz DB. Potentiation of retronasal but not orthonasal in preference learning is modulated selectively by experience. August 2024. *SciFest,* Waltham, MA

10. Goldstein I, Surendran A, Katz DB, Maigler KM, **Gray TR**. Prior Olfactory Exposure Increases Learning in a Cortically Dependent Manner. June 2024. *International Symposium on Olfaction and Taste,* Reykjavik, Iceland

11. **Gray TR,** Goldstein I, Katz DB. Differences in Retro- and Orthonasal Olfaction in Rat Chemosensory Cortices. June 2024. *International Symposium on Olfaction and Taste,* Reykjavik, Iceland

12. **Gray TR,** Goldstein I, Katz DB. Early Exposure to Multisensory Stimuli Potentiates Odor Preference in a Modality Specific Manner. June 2024. *Rosbash Lab 50th Anniversary Alumni Symposium*, Waltham, MA

13. Goldstein I, **Gray TR**, Katz DB. Brief Access Task Approach to Assess Combined Olfactory Preference Learning in Rats. August 2023. *SciFest,* Waltham, MA

14. Svedberg DA, **Gray TR**, Katz DB. Cortical Taste Processing Rapidly Accelerates with Novel Taste Exposures. April 2023. *ACHEMS Annual Conference,* Fort Myers, FL.

15. **Gray TR,** Katz DB. Comparison of Retro- and Orthonasal Olfaction in Rat Cortical Networks. November 2022. *SFN Conference,* San Diego, CA

16. **Gray TR**, Katz DB. The Multisensory Coding of Flavor in Freely Moving Rats. October 2022. *SACNAS NDiSTEM Conference,* San Juan, PR

17. **Gray TR**, Baas-Thomas N, Svedberg DA, Lin J, Katz DB. Simultaneous Recordings in Gustatory and Olfactory Cortex During Retronasal Olfaction. September 2021. *Volen Center for Complex Systems Retreat*, West Dennis, MA.

18. **Gray TR**, Himmel HJ, Letcher JM, Benson MN, Cox DN. Cold nocifensive behavior in various drosophilid species. November 2019. *oSTEM National Conference,* Detroit, MI.

19. **Gray TR**, Himmel NJ, Cox DN. The role of neuropeptides in *Drosophila melanogaster* multimodal sensory neurons in driving stimulus-appropriate behavior. October 2019. *Undergraduate STEM Research Society Seminar Series,* Atlanta, GA.

20. Himmel NJ, Letcher JM, Sakurai A, **Gray TR**, Benson MN, Cox DN. Menthol elicits *Trpm*- and *TrpA1*-dependent rolling in *Drosophila* larvae, suggesting Precambrian origins for TRP-dependent menthol sensing. October 2019. *Platform presentation at biennial Neurobiology of Drosophila meeting, Cold Spring Harbor Laboratories,* Cold Spring Harbor, NY.

21. **Gray TR**, Himmel HJ, Letcher JM, Benson MN, Cox DN. A characterization of cold nocifensive behavior in varied drosophilid species. August 2019. *Annual CASA Summer Research Symposium,* Atlanta, GA.

22. Himmel NJ, Letcher JM, **Gray TR**, Benson MN, Cox DN. TRP-dependent chemical sensing: the Precambrian hypothesis. May 2019. *Poster presented at Annual GSU Brains & Behavior Retreat*, Atlanta, GA.

23. Benson MN, Himmel NJ, Letcher JM, **Gray TR**, Cox DN. The ancient origins of chemosensation. April 2019. *Georgia State University Undergraduate Research Conference*, Atlanta, GA.

24. Himmel NJ, Letcher JM, **Gray TR**, Benson MN, Cox DN. Menthol elicits a *Trpm*- and *TrpA1*-dependent nocifensive response in Drosophila melanogaster larvae. February 2019. *Theo Murphy International Scientific Meeting on the evolution of mechanisms and behaviour important for pain, hosted by The Royal Society*, Chicheley, Buckinghamashire, UK.

25. **Gray TR**, Himmel NJ, Letcher JM, Cox DN. Closely related drosophilid species differ in their behavioral response toward noxious cold. September 2018. *GSU IMSD Poster Session*, Atlanta, GA.

26. **Gray TR**, Himmel NJ, Letcher JM, Cox DN. The evolution of cold nocifensive behavior in drosophilid larvae. September 2018. *Herty Medal Undergraduate Research Symposium*, Lawrenceville, GA.

27. Himmel NJ, Letcher JM, **Gray TR**, Cox DN. The evolution of cold nociception in drosophilid larvae. May 2018. *Annual GSU B&B/IMSD Summer Poster Session*, Atlanta, GA.

28. Himmel NJ, **Gray TR**, Cox DN. Calcium-activated chloride channels are required for distinguishing between noxious and innocuous stimuli in multimodal sensory neurons. April 2018. *Annual meeting of the Genetics Society of America.* Abstract.

29. **Gray TR**, Himmel NJ, Cox DN. The role of the crustacean cardioactive peptide gene in the regulation of the cold nociceptive response in *Drosophila melanogaster*. July 2017. *Annual GSU B&B/IMSD Summer Poster Session*, Atlanta, GA.

PROFESSIONAL SOCIETIES

Sigma Xi Research Honor Society	2026 - Present
Colegio Colombiano de Neurociencias	2024 - Present
Association for Chemoreception Sciences	2021 - Present
Diverse Brandeis Scholars	2020 – Present
Society for Advancement of Chicanos/Hispanics and Native Americans in Science	2019 - Present
Undergraduate STEM Research Society	2019 - 2020
oSTEM	2019 - 2020
Nu Rho Psi Neuroscience Honor Society	2017 - Present

TEACHING EXPERIENCE

Teaching Assistant for Perception: Human, Animal, and Machine Fall 2024
Brandeis University
Instructor: Robert Sekular, Ph.D.

Teaching Assistant for Introduction to Behavioral Neuroscience Spring 2023/2024
Brandeis University
Instructor: Donald Katz, Ph.D.
(Invited to give lecture on Nociception and Somatosensation)

Teaching Assistant for Introduction to Psychology Fall 2022
Brandeis University
Instructor: James Howard, Ph.D.
(Invited to give lecture on Chemosensory Systems)

Teaching Assistant for Introduction to Cognitive Neuroscience Spring 2022
Brandeis University
Instructor: Teresa Mitchell, Ph.D.
(Invited to give lecture on Chemosensory Systems)

Teaching Assistant for General Biology Laboratory Fall 2021
Brandeis University
Instructor: Melissa Kosinski-Collins, Ph.D.

Teaching Assistant for Crustacean Neuroscience Laboratory Spring 2020
Georgia State University
Instructor: Manfred Schmidt, Ph.D.

Teaching Assistant for Fruit Fly Neurogenetics Laboratory Spring 2020
Georgia State University
Instructor: Sarah Clark, Ph.D.

Teaching Assistant for Optogenetics Laboratory July 2019
The Neuroscience School, Georgia State University
Instructor: Sarah Clark, Ph.D.

MENTORING EXPERIENCE

Graduate Student Mentor August 2024 – Present
Master's Student at Brandeis University
Mentee: Ainsley Craddock

Graduate Student Mentor August 2024 – November 2024
Doctoral Student at Brandeis University
Mentee: Jason Lê

HHMI Galaxy Program Mentor August 2022 – May 2024
Undergraduate Student Cohort at Brandeis University
Mentees: 11 first years in 2022, 8 first years in 2023, 5 first years in 2024

Graduate Student Mentor April 2022 – May 2025
Undergraduate Student at Brandeis University
Mentee: Isaac Goldstein Class of 2025
Presentations: SciFest 2023, ISOT 2024, SfN 2024

Peer Graduate Student Mentor September 2021 – May 2022
First year graduate Student at Brandeis University
Mentee: Michael Vivian

***Nu Rho Psi* Graduate Student Mentor** September 2020 – December 2021
Undergraduate Student at Georgia State University
Mentee: Jimena Pamela Castillo

LEADERSHIP POSITIONS

Brandeis University Graduate Student Social Club - President	2021 - 2024
Brandeis University Diverse Brandeis Scholars - Public Relations Chair	2020 - 2023
Georgia State University Rowing Club - President	2018 – 2019
Georgia State University *Nu Rho Psi* - Public Relations Director	2019 - 2020
Georgia State University Out in STEM - Vice President	2019 - 2020
Georgia Senior Classical League - Historian	2016 - 2017

SCIENCE COMMUNITY ENGAGEMENT

Light Sheet Microscope Demonstrator for Brandeis Undergrad Admitted Students Day	March 2026
Waltham High School (12th Grade) Lab Field Trip	February 2026
Advanced Degree Panel Member for GSU Center for Alumni and Student Advancement	July 2025
English Immersion for Spanish Speakers Waltham HS (9-12th Grade) Lab Field Trip	April 2024
Spanish Spoken Research Presentation at Waltham HS (9-12th Grade)	February 2024

NCUR Abstract Reviewer	January 2024
Mentor Judge for Undergraduate Posters at SACNAS 2023	October 2023
SACNAS 2023 Brandeis University Graduate Student Representative	October 2023
Presented Research and Immersive Activity for Waltham Boys & Girls Club	August 2023
Workshop on Personal Branding in STEM for Brandeis 1st year undergraduates	April 2023
English Immersion for Spanish Speakers Waltham HS (9-12th Grade) Lab Field Trip	April 2023
Lead Host for Invited Speaker Keith Hengen, Ph.D.	March 2023
Invited Speaker at Weston HS (9-12th Grade)	March 2023
SACNAS 2022 Brandeis University Graduate Student Representative	October 2022
Lead Host for Invited Speaker Graeme Davis, Ph.D.	May 2022
GSU Cox Lab Alumni Panel Member	March 2022
El Hora de Saber Talk at El Bosque Universidad, Bogota, Colombia (Zoom)	March 2022
Invited Speaker for Science Pizza Talk at Waltham HS (9-12th Grade)	March 2022
"How Your Brain Processes Tastes and Smells" Talk - High Meadows School (4th Grade)	September 2021
SACNAS 2021 Brandeis University Grad Student Representative	October 2021
Going the Distance 2021 Internship Panel Member	June 2021
ABRCMS 2020 Brandeis University Grad Student Representative	November 2020
SACNAS 2020 Brandeis University Grad Student Representative	October 2020
Nu Rho Psi Graduate Student Panel	October 2020
Collegiate Neuroscience Society Career Panel	September 2020
Atlanta Regional Brain Bee Optogenetics Demo Booth	February 2020
CNS Touch a Brain Event at Fernbank Museum of Natural History	February 2020
Science Night "Build-a-Brain" Activity at Ashford Park Elementary	October 2019
oSTEM CV and Resume Workshop Leader	October 2019
Nu Rho Psi Touch a Brain Science Event	August 2019
Center for Advancement of Students and Alumni Panel Member	May 2019
Atlanta Science Festival Station Leader	March 2018
S-Stem Scholars Application Panel Member	August 2017
NCR Scholars Application Panel Member	August 2017

TECHNICAL SKILLS

MOLECULAR BIOLOGY AND NEUROSCIENCE:

BEHAVIORAL SCREENS
DROSOPHILA NEUROGENETICS
IMMUNOHISTOCHEMISTRY
RNA EXTRACTION AND QPCR
CONFOCAL MICROSCOPY
LIGHT SHEET MICROSCOPY
ELECTROPHYSIOLOGY
OPTOGENETICS
TISSUE CLEARING WITH CLARITY
STEREOTAXIC SURGERY
RAT HUSBANDRY

PROGRAMMING AND DATA ANALYSIS:

PYTHON (INTERMEDIATE)
MATLAB (BEGINNER)
FIJI-BASED IMAGE ANALYSIS

IMARIS

ENGINEERING AND TECHNOLOGY:

ELECTRONIC CIRCUIT DESIGN
SOLIDWORKS (CSWA CERTIFIED)
3D PRINTING (STEREOLITHOGRAPHY AND FUSED DEPOSITION MODELING)
MICROCONTROLLER PROGRAMMING
IOT AND EMBEDDED DEVELOPMENT
AIR / LIQUID DISPENSING SYSTEMS
NVIDIA JETSON AI SPECIALIST

HOBBIES

Running

- **Grad PR** Martha's Vineyard Marathon 3:29:03 [8/35 in M 18-24] [39/199 M all] (2022)
- **Grad PR** Cambridge Half-Marathon 1:41:26 [255/604 in M 18-29] [464/1390 in M all] (2021)
- **Grad PR** B.A.A. 5K 0:19:43 [24/114 in M 20-24] [282/3601 M all] (2022)
- **College PR** Kirkwood Spring Fling 5K 21:41 [12/239 in M] [14/539] (2018)
- **HS PR** NGHS CC 5K 21:41 (2016)

Cycling

- L'Enfer du Nord - Weymouth Road Race 14/22 (2022) 7 points
- Zoom Criterium 16/21 (2022) 9 points
- Mooz Criterium 7/19 (2022) 14 points
- URI Cycling Weekend – Westerly Road Race 68/74 (2023)
- URI Cycing Weekend – Charlestown Criterium 13/16 (2023) 9 points
- Pioneer Valley Showdown – Warwick Road Race 70/82 (2023)
- ECCC Road Championship – Frat Row Criterium 7/13 (2023)